

$SU(2)_L$ Chirality from the Galois Structure:

Su/Sd Separation and $Cl(6)$ Grade Parity

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$SU(2)_L$ Chirality from the Galois Structure: Su/Sd Split and $Cl(6)$ Grade Parity

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Status markers

[B] *Established* — secured by script.

Abstract

The $SU(2)_L$ chirality structure of the Standard Model follows algebraically from the $Cl(6)$ grade parity established in Doc. 341. The pseudoscalar $\gamma_5 = e_1 e_2 e_3 e_4 e_5 e_6$ has eigenvalue $(-1)^r$ on grade- r elements. The Su/Sd split (Doc. 344) is therefore exactly the right/left split: Su = even grades = right-handed **[B]**, Sd = odd grades = left-handed **[B]**. The open question from Docs. 347/348 is thereby closed **[B]**.

1 Theorem A — Grade parity and γ_5 eigenvalue [B]

$\gamma_5 \cdot X = (-1)^r X$ for grade- r element $X \in Cl(6)$. Even grades: $\gamma_5 = +1$ (right-handed). Odd grades: $\gamma_5 = -1$ (left-handed). **[B]**

2 Theorem B — Su = even grades = right-handed [B]

Su[0]= v : grade 0 $\rightarrow$ right. Su[1]= $\bar{u}_d$: grade 2 $\rightarrow$ right. Su[2]= u : grade 4 $\rightarrow$ right. Su[3]= e_R : grade 6 $\rightarrow$ right. All $\gamma_5 = +1$. **[B]**

3 Theorem C — Sd = odd grades = left-handed [B]

Sd[0]= $\bar{v}$: grade 1 $\rightarrow$ left. Sd[1]= d : grade 3 $\rightarrow$ left. Sd[2]= $\bar{d}$: grade 5 $\rightarrow$ left. Sd[3]= e_L : grade 7 $\rightarrow$ left. All $\gamma_5 = -1$. **[B]**

4 Theorem D — $SU(2)_L$ doublets from Sd sector [B]

Left-handed doublets: $(v_L, e_L) = (Sd[0], Sd[3])$; $(u_L, d_L) = (Sd\text{-quarks})$. Right-handed singlets: $e_R = Su[3]$; $u_R = Su[2]$; $v_R = Su[0]$. $SU(2)_L$ acts on the Sd sector (left-handed) only. [B]

5 Theorem E — $\mathbb{Z}_3$ fixed points/orbit = chirality [B]

From Doc. 341: $Vss[0, 2, 4]$ (σ -fixed points) = right-handed; $Vss[1, 3, 5]$ (σ -three-orbit) = left-handed. The $SU(2)_L$ chirality split is a direct consequence of the $\mathbb{Z}_3$ symmetry of $T^4/\mathbb{Z}_3$. [B]

6 Summary

Theorem	Content	Status
A	γ_5 eigenvalue $(-1)^r$ on grade r	[B]
B	Su = even grades = right-handed	[B]
C	Sd = odd grades = left-handed	[B]
D	$SU(2)_L$ doublets = Sd sector	[B]
E	$\mathbb{Z}_3$ fixed points/orbit = right/left	[B]

The $SU(2)_L$ chirality structure is now derived from the Galois/Clifford algebra [B]. Open: exact CKM mixing angles and CP phase δ_{CP} (Doc. 348) [S].

Verification script

pruef_349_chiralitaet.py — Theorems A–E, all assertions passed [B].

Note on AI assistance

Prepared with support from Claude (Anthropic). Physical interpretations: Johann Pascher (ORCID 0009-0000-6518-4064).

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